Multimodal Reasoning Benchmarks

Questions ⋯



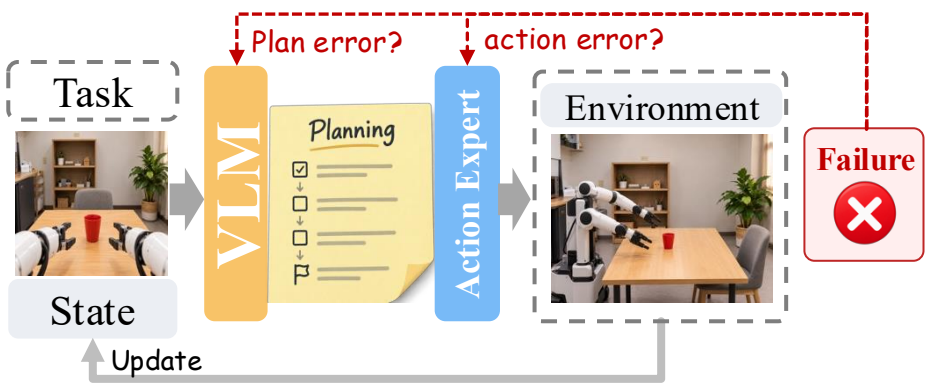
VLM

Reasoning Answer

Answer: She finished the task successfully, as her final celebration follows a sequence of focused work.
…

Limitation: Only evaluates static VLM capabilities

Long-horizon Embodied Benchmarks



Limitation: Reasoning and execution are not decoupled

Gap

Can Existing VLMs Go Well Beyond Perception to ***Reason and Plan*** *in Embodied Worlds?*



Isolate **high-level** reasoning

Remove **execution** confounds

PLAN-Bench

Plan-Bench (Ours)

An execution-decoupled benchmark for evaluating high-level reasoning and long-horizon planning of VLMs in embodied worlds.

Spatial Cognition

multi-view · reflection · spatial cues



A.Purple Green B.Blue C.Purple White D.Blue Purple ✓

Question
Identify empty cups across multiple views and output their colors.

Temporal Reasoning

ordering · tracking · persistence

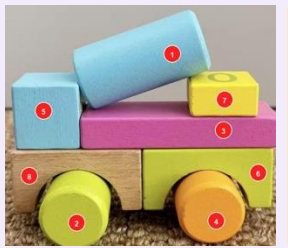


A. 6,5,4,3
B. 5,6,4,3
C. 4,3,6,5 ✓
D. 4,5,3,6

Question
Order images 3–6 based on the event sequence and select the correct option.

Multi-Step Planning

constraints · dependencies · planning



Question
Plan the valid removal order to reach object 6.
✓ 1,7,3,6



Question
Plan the route from D to C via A, F, and B.
✓ 5,5,1,1,3,3,4